

Health-Care No-Show Analysis

A data-driven investigation into the factors driving missed appointments — enabling healthcare administrators to reduce waste, improve scheduling, and strengthen patient engagement.

HEALTHCARE OPERATIONS

PREDICTIVE ANALYTICS



The Operational Cost of Missed Appointments



Why This Matters

When patients miss appointments without notice, clinics face wasted slots, idle staff, longer waits, and lost revenue. This analysis examines demographics, scheduling patterns, reminders, distance, weather, and satisfaction to identify drivers of no-show behavior.

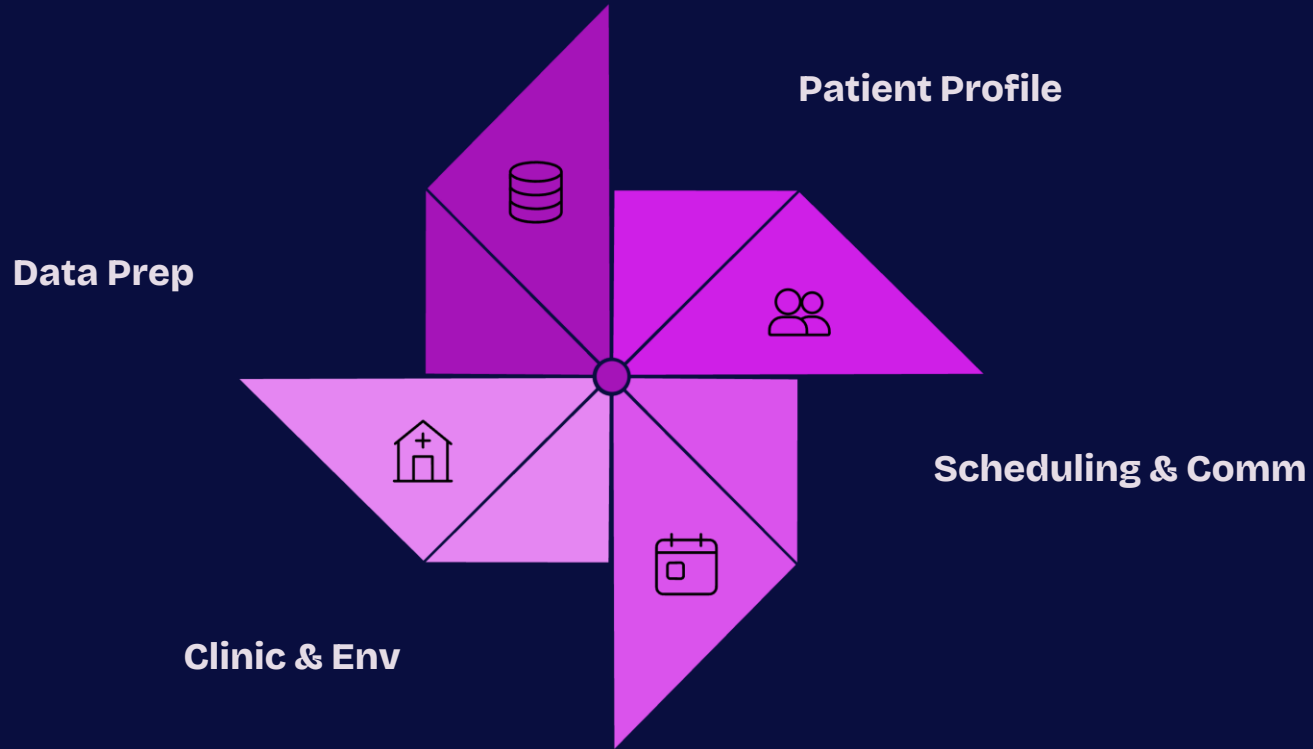
Wasted Slots

Higher Costs

Longer Waits

Lost Revenue

Analysis Roadmap



Each phase isolates a distinct dimension of no-show behavior, building toward actionable, evidence-based recommendations for clinic operations.

Overall No-Show Rate: ~8.3%

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Demographic Variation

Small

Age 8.04%–8.52%

City 8.01%–8.82%

Key Insight

The overall no-show rate is 8.3%. Demographic variables such as age, gender, city, and neighborhood show only minor differences in no-show rates, whereas operational, behavioral, and socioeconomic factors exhibit much stronger associations with patient attendance.

Strongest Indicators

- Waiting Days
- Previous No-Shows
- Sms Message
- Reminder Calls
- Patient Satisfaction
- Income Level
- Previous No-Shows
- Travel Distance
- Insurance Type
- Weather Condition

Dataset Summary

Total Appointments: 50,000 Rows

Overall Attendance:

Show: 91.7%

No-Show: 8.3%

Variables Analyzed: 27 Columns

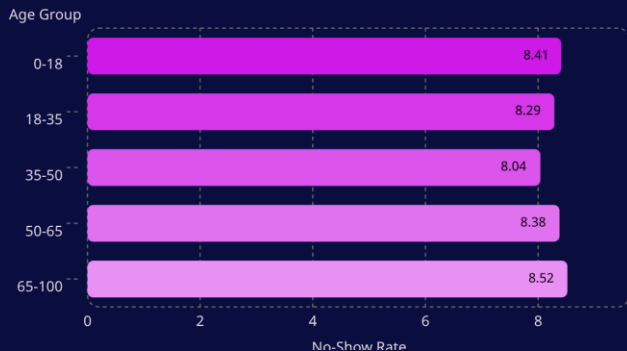
Strong / Very Strong Indicators: 8

Weak / Very Weak Indicators: 11

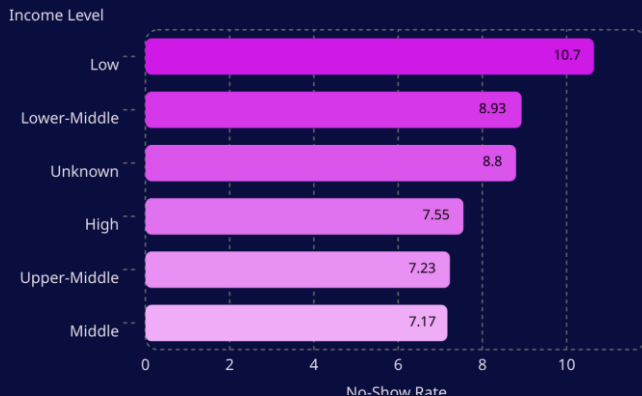
Executive overview of the analyzed dataset.

Patient Demographic Analysis

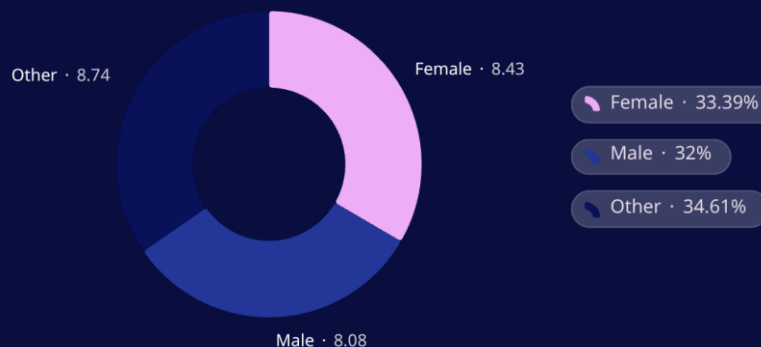
Age Group vs No-Show Rate



Income Level vs No-Show Rate



Gender vs No-Show Rate

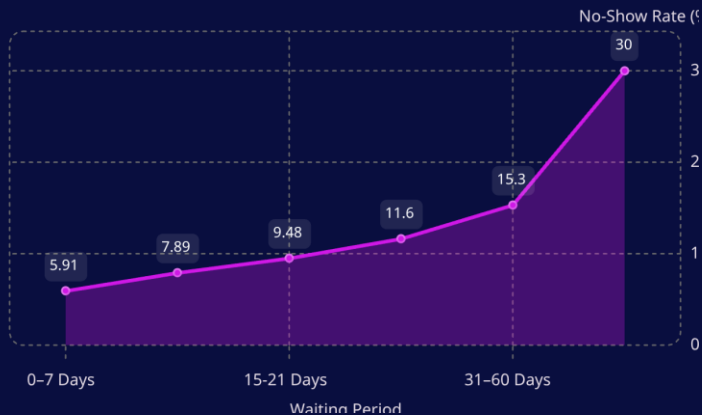


Key Findings

- Age: Very Weak (8.04%-8.52%)
- Gender: Very Weak (8.08%-8.74%)
- Income: Strong (7.17%-10.65%)

Income level shows the strongest demographic relationship with no-show behavior.

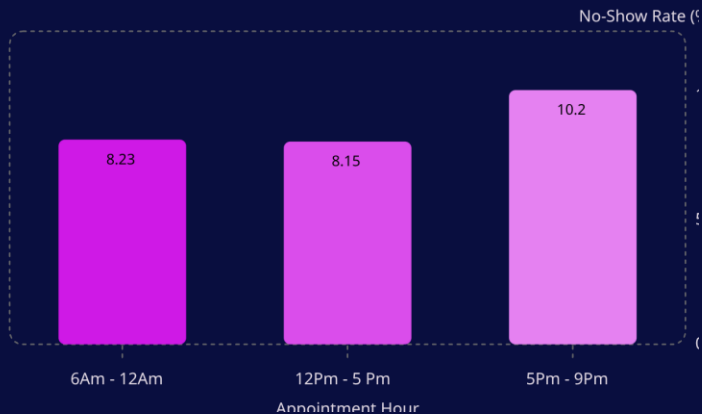
Waiting Time: The Strongest Scheduling Factor



A Clear Escalation Pattern

No-show rates climb from **5.9%** for same-week appointments to **30%** for bookings made over 60 days in advance — a **5× increase**.

Long lead times erode patient commitment. Evening slots (5–9 PM) also show elevated rates at **10.22%** vs. ~8.2% for morning and afternoon.



Reminders & Past Behavior: Powerful Predictors

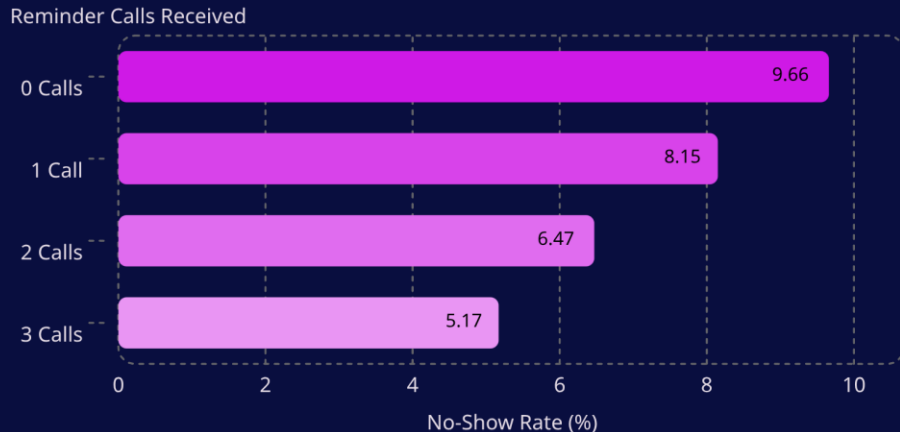


Communication Is Highly Effective

Each additional reminder call reduces no-shows. SMS reminders cut rates from **10.94%** (no SMS) to **7.25%** (SMS sent).

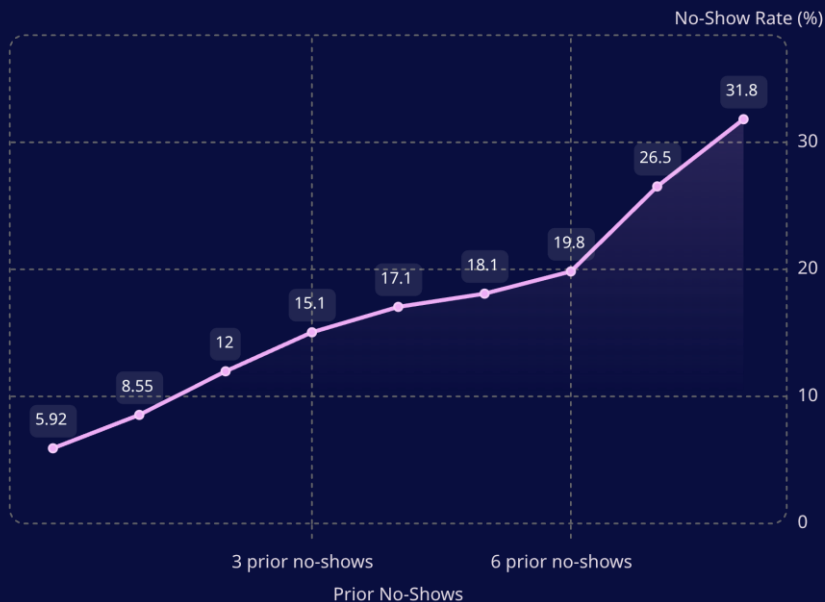
Past Behavior Predicts the Future

Patients with no prior no-shows: **5.92%**. Those with 8+ prior misses: **31.82%** — a **very strong** signal for proactive outreach.



Previous No-Shows: A Very Strong Predictor

No-Show Rate by Prior No-Show History

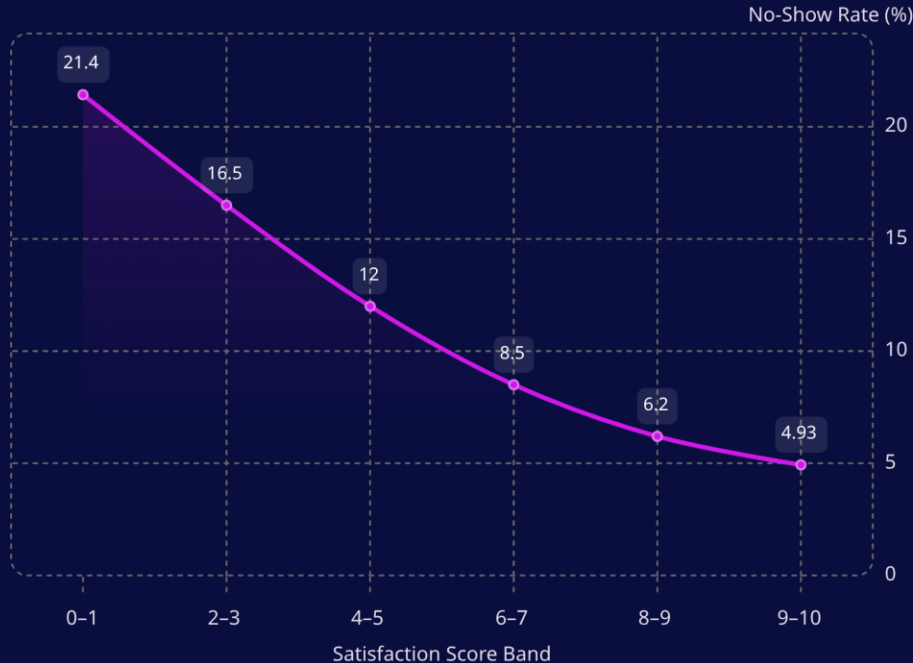


Past Behavior Predicts Future Attendance

A very strong relationship exists between previous no-show history and future behavior. No-show rates escalate from 5.92% (no prior no-shows) to 31.82% (8+ prior no-shows).

This is one of the strongest predictors identified and suggests past no-show behavior is a critical signal for targeted intervention strategies.

Patient Satisfaction: A Strong Inverse Relationship



The Satisfaction–Attendance Link

No-show rates fall from **21.43%** (lowest satisfaction) to **4.93%** (highest) — a **4× difference** and one of the strongest associations in this analysis.

Improving patient experience is not just a quality metric — it is a direct lever for reducing missed visits and optimizing clinic throughput.

Clinic & Environmental Factors



Travel Distance

No-shows rise from **6.86%** (<2 km) to **10.93%** (>20 km). Car users show the lowest rate (7.27%); walkers the highest (9.68%).



Weather Conditions

Rain raises no-shows to **10.43%**; sunny and cloudy days hold rates near **7.4%**. Adverse weather is a meaningful external driver.



Insurance Type

Uninsured patients: **11.04%** no-show rate. Private insurance: **7.60%**. Coverage status is a strong equity-related attendance factor.



Appointment Type

Emergency visits: **4.87%** (lowest). Therapy: **9.02%** (highest). Perceived urgency directly shapes attendance.

Key Takeaways & Recommendations

● Act on Waiting Time

Review scheduling policies for long lead times. Consider additional reminders or follow-up for appointments booked more than 30 days in advance. Cap advance bookings at 30 days. Send automated reminders at 7 and 3 days before the visit.

📞 Scale Reminder Programs

3 calls reduce no-shows to 5.17%. Patients receiving more reminder calls exhibited lower no-show rates. Prioritize reminder programs, particularly for patients with a history of missed appointments

💡 Invest in Satisfaction

Patient satisfaction showed one of the strongest associations with attendance. Improving the patient experience may help support better attendance.

💛 Address Access Barriers

Consider transportation support for patients traveling long distances or uninsured and proactive communication during adverse weather conditions.

